ALL INDIA COUNCIL FOR TECHNICAL EDUCATION · DECISION SUPPORT SYSTEM

Engineering Completion Report: Frontend Freezing Resolution & Professional Scrutiny Report Generator

Project:	**CipherCore Verify (SIH 26261)**	**Status:**	**100% COMPLETE & VERIFIED**
Scope:	Task A (Zero-Freeze) & Task B (ReportLab Generator)	**Date:**	2026-10-27
Automated Tests:	**21/21 Tests Passed (17 DOM / 4 Python)**	**Live Server:**	HTTP 200 OK (Port 3000 Active)

1. TASK A: FRONTEND FREEZING ROOT-CAUSE ANALYSIS & FIX

A rigorous diagnosis identified five interacting defects causing UI unresponsiveness: (1) CSS selector asymmetry where #cvInspectionModal and #aicteScrutinyReportModal were positioned fixed 100vw x 100vh with z-index 999999 but omitted from the hiding selector; (2) Conflicting Tailwind utility classes (hidden + flex) in #uploadDossierModal and #officialDecisionModal; (3) Missing .modal-backdrop class preventing outside click dismissal; (4) Missing overflow cleanup in modal close functions, permanently locking document.body.style.overflow = 'hidden'; (5) Tab navigation desynchronization where switching views did not dismiss open overlays. Pressing Escape was the only key restoring responsiveness because its global listener explicitly called close functions that cleared body overflow.

Architectural Fix Implemented: Created a unified modal lifecycle manager (isAnyModalOpen, checkAndResetBodyOverflow, closeAllActiveModals). Updated switchMainView to auto-dismiss all modals upon tab changes. Enclosed file uploads in try...catch...finally blocks with input reset and scroll unlock. Normalized CSS selectors across all modal IDs to guarantee display: none !important when inactive.

2. TASK B: PROFESSIONAL REPORT GENERATOR (backend/report_generator.py)

Developed a standalone, production-grade Python document generation module at backend/report_generator.py using ReportLab 5.1.0 and Pillow 13.0.0. Eliminated client-side canvas screenshots and window.print() in favor of a vector PDF engine. The generated report features: (1) Official AICTE institutional header; (2) Institutional dossier metadata; (3) Statutory scrutiny executive summary; (4) 6-document scrutiny matrix (Fire NOC, Land Deed, SPPU Affiliation, Faculty PAN, Building Plan, Fee Receipt); (5) YOLOv8 Computer Vision facility inspection (142 vs 150 terminal deficit, coordinates, safety equipment); (6) Forensic land deed raster analysis (ELA 88.4% tamper score); (7) Consortium Proof-of-Authority blockchain audit trail (Merkle root, validator nodes); (8) Digital signature endorsement block. Supports CLI execution, module imports, and structured JSON export.

3. FULL-STACK PROXY & UI INTEGRATION

Updated server.js to handle /api/reports/generate and /api/reports/download by invoking report_generator.py and streaming the resulting binary PDF directly to the client. Maintains backward route parity for /api/validation-runs/:id/report.pdf and .json. Wired the Dossier Inspection toolbar and #reportModal to trigger downloadOfficialPdfReport() with live status banners.

4. AUTOMATED VERIFICATION AUDIT

Test Suite	Test Target	Result
test_freeze_prevention.js (JSDOM)	Modal lifecycle, tab switch dismiss, click outside, overflow unlock	**17/17 PASSED**
test_report_generator.py (Python)	CLI PDF/JSON generation, custom JSON injection, module import	**4/4 PASSED**
HTTP Live Verification	GET /portal.html, GET /api/reports/generate?format=pdf	**HTTP 200 OK**